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Nuclear Power in France

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France derives about 70% of its electricity from nuclear energy, due to a long-standing policy based on energy security.

Government policy, set under a former administration in 2014, aimed to reduce nuclear's share of electricity generation to 50% by 2025. This target was delayed in 2019 to 2035, before being abandoned in 2023.

In February 2022 France announced plans to build six new reactors and to consider building a further eight.

France is the world's largest net exporter of electricity due to its very low cost of generation, and gains over €3 billion per year from this.

The country has been very active in developing nuclear technology. Reactors and especially fuel products and services have been a significant export.

About 17% of France's electricity is from recycled nuclear fuel.

57



OPERABLE
REACTORS

63,000 MWe

0



REACTORS UNDER
CONSTRUCTION

0 MWe

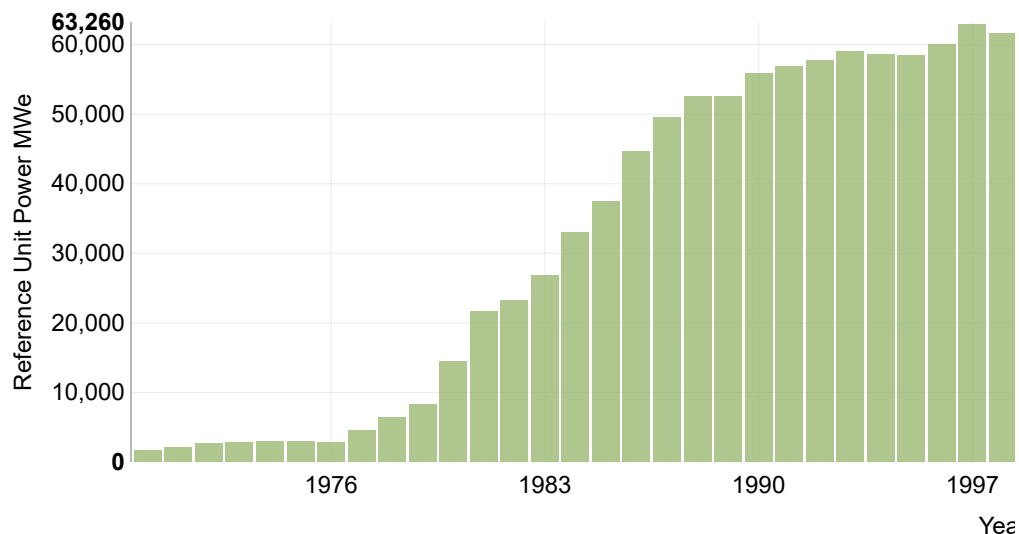
14



REACTORS
SHUTDOWN

5,563 MWe

Operable nuclear power capacity



Electricity sector

Total generation (in 2024): 570 TWh

Generation mix: nuclear 380 TWh (67%); hydro 77.5 TWh (14%); wind 47.2 TWh (8%); solar 23.6 TWh (4%); natural gas 17.7 TWh (3%); biofuels & waste 12.2 TWh (2%); oil 7.0 TWh; coal 2.0 TWh.

Import/export balance: 89.9 TWh net export (15.1 TWh imports; 105 TWh exports)

Total consumption: c. 400 TWh

Per capita consumption: c. 5800 kWh in 2024

Source: International Energy Agency and The World Bank. Data for year 2024.

Installed capacity at the end of 2024 was 156 GWe.

In 2024 France was Europe's biggest electricity exporter.

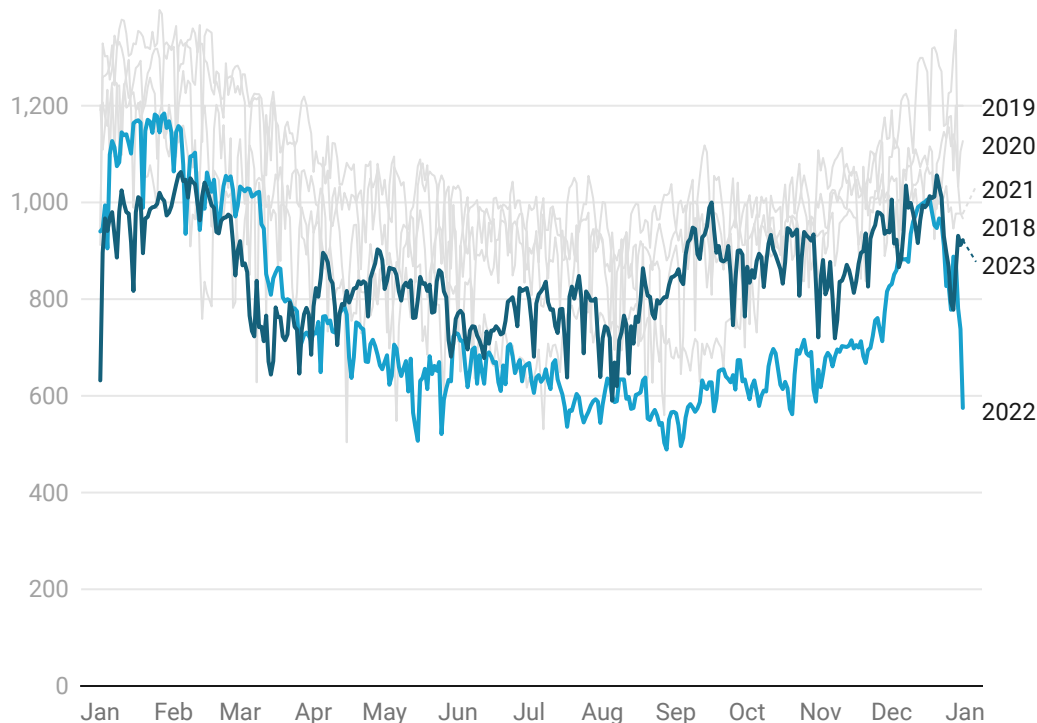
France's present electricity generation mix is a result of the French government deciding in 1974, just after the first oil shock, to rapidly expand the country's nuclear power capacity, using Westinghouse technology. This decision was taken in the context of France having substantial heavy engineering expertise but few known indigenous energy resources. Nuclear energy, with the fuel cost being a relatively small part of the overall cost, made good sense in minimizing imports and achieving greater energy security.

As a result of the 1974 decision, France now claims a substantial level of energy independence and an extremely low level of carbon dioxide emissions per capita from electricity generation, since over 80% of its electricity is from nuclear or hydro.

Nuclear outages in 2022

In 2022 France's reactor fleet produced 282 TWh, well below the ten-year average of 395 TWh. Output rose in 2023 to 320 TWh as reactors returned to service following inspections and repairs. Nuclear output was 373 TWh in 2025. EDF estimates output of 350-370 TWh in 2026.

Reduced output in 2022 was due to: repairs for or investigations of corrosion issues that were first detected at Civaux 1 in December 2021 (for more information [see below](#)); the knock-on effect of planned outages being delayed or reduced in scope in 2021 due to the Covid-19 pandemic; and industrial action. EDF recorded a record net loss of €17.9 billion in 2022.



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France daily nuclear electricity production, MWh (2016-2023).
2020 production lower due to Covid lockdowns (source: Open Data Réseaux Énergies)

Stress corrosion and outages

In December 2021 maintenance checks on the primary circuit of Civaux 1 revealed corrosion near the welds on pipes of the safety injection system. Checks were then carried out on the same equipment at unit 2, revealing similar defects. EDF decided to replace the affected parts, requiring an extended shutdown of the

plant, and also take its two other N4 units at Chooz B offline to carry out similar checks.

In mid-January 2022, EDF announced that similar faults on the safety injection system pipe welds to those discovered at Civaux 1 had been found at Civaux 2 and Chooz B2. In addition, the ten-year in-service inspection at Penly 1 – one of twelve 1300 MWe-class units of the P'4 series – also revealed stress corrosion. (The piping of the P'4 units differs from that of the eight older P4 series units.)

France's regulator, ASN, said that the issue of corrosion was likely due to the “geometry of the lines” in EDF's N4 and P'4 reactors. According to the regulator, the layout of lines promotes a thermal stratification of fluid in the affected pipes, which generates stress in the weld zones.

EDF has identified the pipework most susceptible to cracking:

N4 reactors – stress corrosion in the safety injection circuit located in the 'cold leg' (i.e. the pipes of the main primary circuit which go from the motor pump units to the reactor vessel) and in the pump lines of the shutdown reactor cooling circuit.

P'4 reactors – stress corrosion limited to the injection circuit in the cold leg.

ASN in July 2022 said: “Based on the available knowledge, it would appear that the susceptibility of the reactors of the P4 and 900 MWe plant series to the stress corrosion phenomenon is low to very low.”

A green bilateral loan worth €1 billion was announced by EDF and Credit Agricole CIB in November 2022. The loan will be used to finance maintenance works across the French nuclear power fleet and is part of EDF's Grand Carenage program, which aims to improve the safety and continue the operation of France's nuclear fleet beyond 40 years.

In March 2023 EDF presented an evolution to the stress corrosion control strategy to ASN. Following a laboratory examination of the cracks discovered in Unit 1 of the Penly nuclear power plant, EDF believes that an initial repair job on the affected weld likely contributed to a significant increase in the local residual stresses, giving the appearance of a stress corrosion defect.

EDF completed the inspection and repair of the 12 reactors (four N4 and eight P'4 units) considered most sensitive to stress corrosion by January 2024 (see below).

EDF has been inspecting all of its reactors, prioritizing the N4 and P'4 units, with inspections continuing into 2026.

Reactor	Reactor class	Findings	Restart date
Belleville 1	P'4	SC found; repairs completed	Dec 2023
Belleville 2	P'4	SC found; repairs completed	Sep 2023
Cattenom 1	P'4	SC found; repairs completed	Aug 2023
Cattenom 2	P'4	SC found; repairs completed	July 2023
Cattenom 3	P'4	SC found; repairs completed	April 2023
Cattenom 4	P'4	SC found; repairs completed	Dec 2022
Golfech 1	P'4	SC found; repairs completed	Jan 2024
Golfech 2	P'4	SC found; repairs completed	Sep 2023
Nogent 2	P'4	SC found; repairs completed	July 2023
Penly 1	P'4	SC found; repairs completed	July 2023
Penly 2	P'4	SC found; repairs completed	June 2023
Chooz B1	N4	SC found; repairs completed	May 2023
Chooz B2	N4	SC found; repairs completed	Feb 2023
Civaux 1	N4	SC found; repairs completed	Jan 2023
Civaux 2	N4	SC found; repairs completed	Apr 2023

Energy policy

In 1999 a parliamentary debate reaffirmed three main planks of French energy policy: security of supply, respect for the environment (especially re greenhouse gases) and proper attention to radioactive waste management. It was noted that natural gas had no economic advantage over nuclear for base-load power, and its prices were very volatile. It was accepted that there was no way renewables and energy conservation measures could replace nuclear energy in the foreseeable future.

Early in 2003 France's first national energy debate was announced, in response to a "strong demand from the French people", 70% of whom had identified themselves as being poorly informed on energy questions. A poll had shown that 67% of people thought that environmental protection was the single most important energy policy goal. (However, 58% thought that nuclear power caused climate change while only 46% thought that coal burning did so). The debate was to prepare the way for defining the energy mix for the next 30 years in the context of sustainable development at a European and at a global level.

In 2005 a law established guidelines for energy policy and security. The role of nuclear power was central to this, along with specific decisions concerning the European Pressurised Water Reactor (EPR), notably to build an initial unit so as to be able to decide by 2015 on building a series of about 40 of them. It also set out research policy for developing innovative energy technologies consistent with reducing carbon dioxide emissions and it defined the role of renewable energies in the production of electricity, in thermal uses and transport.

Early in 2008 a Presidential decree established a top-level **Nuclear Policy Council** (*Conseil Politique Nucléaire – CPN*), underlining the importance of nuclear technologies to France in terms of economic strength, notably power supply. It is chaired by the President and includes the prime minister as well as the cabinet secretaries in charge of energy, foreign affairs, economy, industry, foreign trade, research and finance. The head of the Atomic Energy Commission (CEA), the secretary general of national defence and the military chief of staff are on the council. (See section below on Nuclear technology exports for further information.)

Following the election of President Francois Hollande in 2012 with his policy to reduce the proportion of nuclear power in the energy mix, a new wide 'national debate on energy transition' was called, which ran eight months to July 2013. The Ministry for Ecology Sustainable Development and Energy counted 170,000 people taking part in 1000 regional debates, and received 1200 submissions over the Internet. A report published in September 2013 by OPECST, a scientific commission of senators and MPs from the upper and lower houses of Parliament said France risked being exposed to a power price shock if it pursued a speedy reduction of nuclear power and there was insufficient replacement through renewable energy and energy efficiency measures.

In October 2014 the **Energy Transition for Green Growth bill** was passed by the National Assembly and so went onto the Senate. This set a target of 50% for nuclear contribution to electricity supply by 2025, and capped nuclear power capacity at 63.2 GWe, the level at the time. This meant that EDF would have needed to shut at least 1650 GWe of nuclear capacity when its Flamanville 3 EPR started commercial operation. The bill also set long-term targets to reduce greenhouse gas emissions by 40% by 2030 compared with 1990 levels, and by 75% by 2050; to halve final energy consumption by 2050 compared with 2012 levels; to reduce fossil fuel consumption by 30% by 2030 relative to 2012; and to increase the share of renewables in final energy consumption to 32% by 2030. The Senate early in 2015 amended the bill to remove the nuclear cap, but this was not accepted in the lower house. The National Assembly approved the bill including 970 amendments in July 2015, but with the 63.2 GWe nuclear cap and only 50% nuclear supply by 2025.

In October 2016 the government postponed until after the 2017 presidential and National Assembly elections any decision on which, if any, reactors would close in order to reduce the nuclear share to 50%. In 2017 France postponed its 2025 target for reducing the share of nuclear to 50%. In December 2017 the French President stated that nuclear is "the most carbon-free way to produce electricity with renewables." In November 2018, a draft of the country's new energy plan confirmed that 2035 was the new target date for the reduction of nuclear's share to 50%. The plan states that 14 of the country's nuclear reactors will shut down by 2035, 4-6 of those by 2030. However the plan also states that the option to build new nuclear reactors remains.

A government consultation document released in January 2020 named Blayais, Bugey, Chinon, Cruas, Dampierre, Gravelines, and Tricastin as the plants where EDF planned to make closures to meet the government's target. The document stated that a decision on early shutdowns would be made in 2023, and that following Fessenheim in 2020, the next plant closures were expected 2027-2028.

In November 2021 the French President announced that France was preparing to start construction of new reactors. In January 2022 the Minister for the Ecological Transition said that plans for new reactors were to be submitted around 2023 with a target date of 2035-7 for the reactors to be commissioned. The new reactors are to be EPR2 models.

Earlier in March 2016 Areva, EDF and CEA announced the formation of the tripartite French Nuclear Platform (PFN) to improve the joint effectiveness of the three bodies and devise a shared vision of a medium- and long-term goal for the industry, supporting the Nuclear Policy Council (CPN). Its initial agenda will include the review of technological options for the EPR NM reactor design and the coordination of positions on regulatory changes, notably regarding safety requirements and objectives. The PFN will also address the future of reprocessing in France and elsewhere, the CIGEO deep geological repository project, the development of dismantling technologies for decommissioned reactors, and R&D work on fourth-generation reactor designs.

In October 2019 the environment and economy ministers asked EDF to study the potential for building three pairs of EPR2 reactors at three existing nuclear sites in France. They initially planned to make a decision by mid-2021 on a possible programme for building such capacity, in line with its January 2019 energy plan (PPE), but delayed doing so whilst construction continued at Flamanville 3. The possible new reactor programme relates to a submission to the government by SFEN, the French Nuclear Energy Society, urging such a programme and saying that the construction cost of new reactors could be reduced by 30% and their financing cost by 50%.

In February 2022 President Emmanuel Macron announced plans to build six new reactors, and to consider building a further eight. The President highlighted the need to increase electricity supply by “up to 60%” as the country attempts to reduce consumption of oil and gas over the next 30 years. Macron stated: “Key to producing this electricity in the most carbon-free, safest and most sovereign way is precisely to have a plural strategy... to develop both renewable and nuclear energies. We have no other choice but to bet on these two pillars at the same time. It is the most relevant choice from an ecological point of view and the most expedient from an economic point of view and finally the least costly from a financial point of view.”

In January 2023 the French Senate approved a draft bill – 239 votes in favour and 16 against – to accelerate procedures relating to the construction on new nuclear facilities and the operation of existing facilities. The bill includes:

- removal of the objective to reduce the nuclear share of France's electricity production to 50% by 2035, instead maintaining it at more than 50% by 2050;

- the possibility of constructing new EPRs and SMRs by 2050;
- an exemption from certain urban planning permissions for the construction of new reactors; and
- the option to use an immediate possession procedure to obtain land on which to build new reactors.

In March 2023 France's Parliament formally approved the government's nuclear investment plan – by 402 votes in favor and 130 against – which considers the €52 billion construction of six new EPR-2 PWRs at three sites. President Macron plans to begin construction of the first reactor before the end of his current term in May 2027.

In February 2026 the French government published its third Multiannual Energy Programme (MEP), setting out the country's energy strategy to address the challenges of energy consumption and production over the period 2026-2035. It calls for the construction of six new reactors and the extended operation of the country's existing nuclear fleet, setting a target for nuclear production of 380-420 TWh/yr between 2030 and 2035. The previous MEP set a range of 360-400 TWh/yr.

ARENH

The ARENH scheme (*Accès Régulé à l'Énergie Nucléaire Historique*, Regulated Access to Historic Nuclear Power), implemented on 1 July 2011, granted alternative suppliers the right to purchase electricity from EDF at a regulated price. This mechanism could also be accessed by network operators for their losses. The scheme was managed by the French Energy Regulatory Commission (*Commission de régulation de l'énergie*, CRE).

The price of the ARENH, which was maintained at €42/MWh after January 2012, was proposed by the CRE and determined by the Minister of Energy and the Minister for the Economy. The maximum ARENH overall volume that could be sold to suppliers was set at 100 TWh per year. In 2020, EDF supplied 100 TWh to alternative suppliers, plus 26.2 TWh to offset losses by network managers.

In January 2022 the government announced an additional ARENH allocation of 20 TWh in 2022 to help limit the rise in electricity prices, taking the scheme to 120 TWh and about one third of EDF's production. This was sold between 1 April and 31 December 2022 to alternative suppliers for €46.2/MWh, with the same volume of

electricity being sold by the suppliers back to EDF during the year at €257/MWh or higher. Later in 2022 EDF filed a lawsuit against the French government for more than €8 billion in lost earnings.

ARENH ended on 31 December 2025.

New financial measures were negotiated in 2023 and introduced in 2026 intended to regulate the price of nuclear electricity at an average of €70/MWh.

Legally defined as tariffs on the use of nuclear fuel, the new tariffs see any electricity sold above a 'taxation threshold' taxed at 50% of the extra revenue generated with this redistributed to end users. If prices exceed a higher 'capping threshold', 90% of the revenue would be taxed and redistributed. These levels are intended to be revised regularly by the CRE. In January 2026 when the scheme began, the taxation threshold was €78/MWh and the capping threshold was €110/MWh.

Nuclear power plants

France has 18 commercial nuclear power plants with a total of 57 operable reactors. All power plants are operated by EDF.



France's nuclear power programme cost some FF 400 billion in 1993 currency*, excluding interest during construction. Half of this

was self-financed by EdF, 8% (FF 32 billion) was invested by the state but discounted in 1981, and 42% (FF 168 billion) was financed by commercial loans. In 1988 medium and long-term debt amounted to FF 233 billion, or 1.8 times EdF's sales revenue. However, by the end of 1998 EdF had reduced this to FF 122 billion, about two-thirds of sales revenue (FF 185 billion) and less than three times annual cash flow. Net interest charges had dropped to FF 7.7 billion (4.16% of sales) by 1998.

* 6.56 FF = €1 (Jan 1999)

From being a net electricity importer through most of the 1970s, France has become the world's largest net electricity exporter, with electricity being the fourth largest export. (Next door is Italy, without any operating nuclear power plants. It is Europe's largest importer of electricity, most coming ultimately from France.) The UK has also become a major customer for French electricity.

France's nuclear reactors comprise 90% of EdF's capacity and hence are used in load-following mode (see section below), so their capacity factor is low by world standards, at about 70%.

Licence renewal and uprates

The average age of EDF's fleet of reactors is about 39 years as of early 2026.

French reactors were originally only licensed to 30 years. They are now subject to ten-year reviews to allow their continued operation.

All of the country's 900 MWe reactors started up in the late 1970s to early 1980s. They are reviewed together in a process that takes four months at each unit. In February 2021 ASN announced its approval of a further ten-year operation period for the country's then 32 operating 900 MWe reactors. Various improvements and measures will be applied during the 10-year period. As of February 2021, required upgrade works had been completed at Tricastin 1 and Bugey 2, with the full programme scheduled to run until 2031.

A review of the 1300 MWe class followed and in October 2006 the regulatory authority cleared all 20 units for an extra ten years' operation conditional upon minor modifications at their 20-year outages over 2005-14. The third ten-year inspections of the 1300 MWe series run from 2015 to 2024.

Earlier in July 2010 EdF said that it was assessing the prospect of 60-year lifetimes for all its existing reactors. This would involve replacement of all steam generators (three in each 900 MWe reactor, four in each 1300 MWe unit) and other refurbishment, costing €400-600 million per unit to take them beyond 40 years. Generally, for CP0 and CP1 versions of the 900 MWe M310 plants, extending operation beyond 50 years is not economic.

In February 2014 EDF gave parliament a breakdown of its €55 billion *grand carénage* reactor life extension program, mostly to be completed by 2025. This includes spending €15 billion replacing heavy components within its fleet of 58 nuclear units, €10 billion on post-Fukushima modifications and €10 billion to boost safety against external events. It pointed out that there are only two parts of a nuclear reactor that cannot be replaced, the reactor pressure vessel and the reactor containment building. The rest of the components have a normal lifespan of 25-35 years and require renovation or replacement. ASN said it would evaluate life extensions on the basis of Generation III criteria regardless of when particular reactors were built. In 2017 EdF's *grand carénage* cost estimate to 2025 was reduced to €48 billion, including both maintenance and upgrading, but in October 2020 it was increased to €49.4 billion.

In March 2015 the ASN said that there were no generic elements to prevent the twenty 1300 MWe units operating safely to 40 years. It considers the actions planned or already taken by EDF to assess the condition of the reactors and control ageing issues up to their fourth inspection are adequate. In July 2025 the ASN similarly concluded its generic analysis of 1300 MWe plants operating beyond 40 years, taking into account EDF's provisions as well as prescribing certain work.

In February 2023 President Macron's office reported that the administration had been given the green light to explore the feasibility of extending the operating lives of its nuclear reactors beyond 60 years, including a thorough examination of the nuclear fuel cycle and associated waste management issues.

In August 2023 the ASN approved the extended operation of Tricastin 1, making it the first reactor in the French fleet to operate beyond 40 years.

Fessenheim shutdown

In 2012 the government announced that both Fessenheim reactors should close by 2017. This was for political reasons and regardless of safety evaluations, and would require compensation payments to minority owners: Germany's EnBW has 17.5% and Alpiq, Axpo and BKW in Switzerland together hold 15%. In September 2014 a parliamentary report was presented to the National Assembly confirming that there were no technical reasons for closing the plant, and closing it in 2016 would cost the state some €5 billion, including some €4 billion in compensation to EDF. It was at the time generating average annual profits of some €200 million and allowing it to continue operating after 2016 until 2040 would result in profits of some €4.7 billion. The report concluded: "Whatever the long-term energy policy followed, it would make sense, fiscally and economically, to retain the benefit of the 'surplus nuclear' by not prematurely closing second generation plants currently in operation." The energy minister said that in the light of recent investment in Fessenheim, maybe some other units would close instead. Then in November 2015 the government agreed to EDF's proposal to close Fessenheim only after Flamanville 3 was fully commissioned. In August 2016 the government agreed to pay EDF compensation for the closure, in two instalments, the precise amounts depending on wholesale electricity prices through to 2041. In January 2017 EDF agreed to the compensation protocol which was to be signed when EDF formally requested shutdown. The initial fixed portion of about €490 million would cover the anticipated costs associated with the closure of Fessenheim. This would include retraining of staff, decommissioning the plant, the basic nuclear facility tax and post-operational costs. Some 20% of this initial payment was to be made in 2019, with the remainder in 2021. Further variable payments were to be made to reflect EDF's operating income shortfall up to 2041 due to the closure.

In April 2017 the EDF board decided to give notice of Fessenheim shutdown within six months prior to full commissioning of Flamanville 3. However, that notice would only be given if "the closure of the Fessenheim power plant is necessary in order to comply with the legal ceiling of 63.2 GW both on the date of the request for repeal and on the date of commissioning Flamanville 3," EDF said. "The decision of the board, taken in application of the law and respecting the company's social interest, enables EDF, fully committed to the energy transition, to have the nuclear fleet necessary to fulfil its obligations to supply its customers." The then French energy minister responded by saying that the French state would "legally enshrine" the "inevitable and irreversible" closure of

Fessenheim, and immediately published a decree saying that EDF's authorization to operate the plant's two reactors will be withdrawn from the day that the Flamanville 3 EPR enters into service. The Council of State later ruled the decree had not been issued at EDF's request, as required by law. France's energy minister stated that the closures of Fessenheim 1&2 were no longer linked to the commissioning of Flamanville 3. Fessenheim 1 was shut down on 22 February 2020, followed by unit 2 in June 2020. In November, EDF restarted four coal-fired plants to meet demand.

Active dismantling at Fessenheim is expected to begin in 2026.

Reactor designs and engineering

The first nine power reactors were gas-cooled UNGG (*uranium naturel graphite gaz*) units, designed by the Atomic Energy Authority ([CEA](#)). They were similar to the British Magnox units but developed independently. (One UNGG unit was built in Spain.) EdF then chose pressurised water reactor (PWR) types, supported by new enrichment capacity and fully indigenous manufacturing. EdF's plans for some BWR units did not proceed.

All French units (the first two derived from US Westinghouse types) are now PWRs of three standard types designed by Framatome (later Areva, now Framatome again): three-loop 900 MWe (32), four-loop 1300 MWe P4 type (20) and finally four-loop 1450 MWe N4 type (4). This is a higher degree of standardization than anywhere else in the world. French development of the four-loop 1300 MWe design flowed back to later US plants, and the 1450 MWe N4 design evolved from it.

Framatome in conjunction with Siemens in Germany then developed the **European Pressurised Water Reactor (EPR)**, based on the French N4 and the German Konvoi types, to meet the European Utility Requirements and also the US EPRI Utility Requirements. This was confirmed in 1995 as the new standard design for France and it received French design approval in 2004.

Areva NP (now Framatome) has been working with EdF on a 'new model' EPR – EPR NM or **EPR2** – with simplified construction and significant cost reduction. EdF has said that it, not the complex EPR built at Flamanville, would be the model that replaces the French fleet from the late 2020s. The EPR2 is specified in the October 2019 government request to EdF for proposals to consider building six new reactors. It estimated that the cost of building the six units

would be at least €56 billion. In September 2019 EdF called for civil engineering bids for a pair of EPR2 reactors at an unspecified existing site, involving 800,000 cubic metres of concrete.

There have been two significant **fast breeder reactors** in France. Near Marcoule is the 233 MWe Phénix reactor, which started operation in 1974 and was jointly owned by the CEA and EdF. It was shut down for modification 1998-2003, returned at 140 MWe for six years, and ceased power generation in March 2009, though it continued in test operation and to maintain research programmes on on waste disposal, particularly transmutation of actinides, by the CEA until October 2009.

A second unit was Superphénix of 1200 MWe, which started up in 1996 but was closed down for political reasons at the end of 1998 and is now being decommissioned.

All but three of EdF's nuclear power plants (12 reactors) are inland, and require fresh water for cooling. Eleven of the 15 inland plants (32 reactors) have cooling towers, using evaporative cooling, the others use simply river or lake water directly. With regulatory constraints on the temperature increase in receiving waters, this means that in very hot summers generation output may be limited.

Following the March 2011 at Japan's Fukushima Daiichi nuclear plant, the IRSN undertook a six-month review of reactor safety. Its report, released in conjunction with the ASN, proposed a new set of 'hard core' safety requirements to ensure the protection of vital safety-critical structures and equipment to maintain vital functions in the face of beyond design basis events, such as earthquakes, fires, or the prolonged loss of power or emergency cooling.

New nuclear capacity

Flamanville 3

In mid-2004 the board of EdF decided in principle to build the first demonstration unit of an expected series of Areva EPRs. This decision was confirmed by the EdF board in May 2006, after public debate, when it approved construction of a new 1650 MWe class EPR unit at **Flamanville**, Normandy, alongside two existing 1300 MWe units. The decision was seen as "an essential step in renewing EDF's nuclear generation mix".

The overnight capital cost or construction cost was expected to be €3.3 billion in 2005 Euros (€4.2 billion in 2020 euros), and power from it 4.6 ¢/kWh. Series production costs were projected at about 20% less. EDF then submitted a construction licence application.

Under a 2005 agreement with EdF, the Italian utility ENEL was to have a 12.5% share in the Flamanville 3 unit, taking rights to 200 MWe of its capacity and being involved in design, construction and operation of it. However, early in 2007 EdF backed away from this and said it would build the plant on its own and take all of the output. Nevertheless, in November 2007 an agreement was signed confirming the 12.5% ENEL investment in Flamanville – estimated to cost €450 million – plus the same share of another five such plants. The agreement also gave EdF an option to participate in construction and operation of future ENEL nuclear power plants in Italy or elsewhere in Europe and the Mediterranean. But in December 2012 ENEL pulled out of the project and partnership with EdF and agreed to be reimbursed €613 million that it had contributed, including accrued interest. ENEL said it would pursue its commercial business in France by other means.

Site works at Flamanville on the Normandy coast were complete and the first concrete was poured in December 2007, with construction to take 54 months and commercial operation expected in May 2012. In January 2007 EdF ordered the nuclear steam supply system from Areva. The turbine-generator section was ordered earlier in 2006 from Alstom. This was intended to ensure that 85% of the unit's projected cost was largely locked in. The reactor vessel nozzle support ring was forged by Japan Steel Works (JSW) in 2006. The reactor pressure vessel (RPV) was manufactured at Areva's Creusot Forge St Marcel factory, with delivery to the site in October 2013 and installation in January 2014. In April 2015 tests showed that parts of the RPV steel from Creusot Forge had a high carbon content and one-third lower than specified toughness, and the head of ASN said that it would make an assessment of the slight carbon heterogeneity. Over the next two years China's National Nuclear Safety Administration (NNSA) was involved with this process, since the steam generators for Taishan 1 are also from Creusot Forge. In June 2017 the ASN in a provisional opinion said that the Flamanville RPV was safe for operation, but that EdF should replace the vessel head by the end of 2024. ASN said it would order additional periodic inspections of the bottom of the RPV. In October 2018 ASN authorized the use of the RPV, subject to the implementation of a thermal ageing monitoring programme and on specific controls during operation.

The authorization is for the vessel only. ASN subsequently allowed the vessel head to remain in use until the end of the reactor's first complete operating cycle, with replacement scheduled during the first planned maintenance outage.

As well as the RPV, forging of steam generator shells was at Areva's Creusot Forge factory from 2007, with installation in 2014. A replacement RPV head was manufactured by Framatome, with delivery in late 2024.

At the end of 2008 the overnight cost estimate (without financing costs) was updated by 21% to €4 billion in 2008 Euros (€2434/kW), and electricity cost to be 5.4 cents/kWh. These costs were confirmed in mid 2009, when EdF had spent nearly €2 billion. In July 2010 EdF revised the overnight cost to about €5 billion and the grid connection to early 2014 – two years behind schedule. In July 2011 EdF again revised the completion time to 2016 due to re-evaluation of civil engineering works and to take into account interruptions during the first half of the year. The cost was then put at €6 billion. In December 2012 EdF raised the cost estimate to €8.5 billion including financing, and said that completion was still expected in 2016. As the reactor pressure vessel was installed in January 2014 Areva confirmed that first power was expected in 2016, four years behind the original schedule. In September 2015 the completion date was moved to late 2018, with the cost increasing to €10.5 billion. In July 2017 EdF said that 98% of the civil structure was completed and 60% of the electro-mechanical work, and that the reactor would be connected to the grid in May 2019. In July 2018 EDF announced that quality discrepancies had been found in welds in the secondary coolant system, and that this would delay commissioning by almost a year, and increase the project cost to €10.9 billion. In March 2020 a government decree put off full commissioning until April 2024. The cost estimate from EDF then was €12.3 billion, with start-up in 2023. In January 2022, fuel loading of the reactor was pushed back from late 2022 to the second quarter of 2023, with the estimated completion cost rising to €12.7 billion. EDF announced a further delay in December 2022, postponing fuel loading by an additional six month to the first quarter of 2024, with the cost at completion increasing by €500 million to €13.2 billion. The delays have been put down to supplementary studies needed to establish a new process for the stress-relieving heat treatment of welds.

EDF completed fuel loading in May 2024. The unit was connected to the grid in December 2024 and reached 100% power in

December 2025.

Penly 3&4

The first of three planned pairs of EPR2 reactors are planned to be constructed at Penly nuclear power plant in Normandy. Site groundworks began in July 2024. Earlier in 2023 EDF and French company Eiffage signed a €4 billion contract for the main civil engineering works at the site, including the construction of the two EPR2 units and 69 associated civil structures.

A public inquiry on the construction licence application ran from 22 January to 4 March 2026. ASN is expected to issue its formal opinion to the government by the end of 2026.

Other proposals

In August 2005 EdF announced that it planned to replace its fleet of reactors with EPR units from 2020, at the rate of about one 1650 MWe unit per year. It would require 40 of these to reach present capacity. This was to be confirmed on the basis of experience with the initial EPR unit at Flamanville – use of other designs such as Westinghouse's AP1000 or GE's ESBWR was considered possible. EdF's development strategy had selected the nuclear replacement option on the basis of nuclear's "economic performance, for the stability of its costs and out of respect for environmental constraints." However, in mid-2015 Areva said that it was working with EdF on a 'new model' EPR – the EPR NM or EPR 2 – with simplified construction and significant (likely 25%) cost reduction. The basic design was 30% complete by March 2016, and EdF said that it, not the complex EPR being built at Flamanville, would be the model that replaced the French fleet from the late 2020s.

In January 2009 President Sarkozy announced that EdF would build a second 1650 MWe EPR, at **Penly**, near Dieppe, in Normandy. Like Flamanville, it has two 1300 MWe units now operating, and room for two more. GdF-Suez originally planned to hold a 25% stake in it, with Total holding 8.3%, and Enel 8-12.5% entitlement. Germany's E.ON was considering taking an 8% stake. A public debate on the project concluded in 2010, but nuclear safety authority ASN did not accept EdF's application to build the unit, sending it back for further work before submission to a local public inquiry. However, EdF then

halted plans for the Penly 3 unit and said that it did not intend to build more nuclear capacity in France for operation before 2025.

A third new reactor, with majority GdF Suez ownership and operated by it, was proposed to follow – in line with the company's announced intentions. A GdF Suez (now Engie) subsidiary, Electrabel, operates reactors in Belgium and has equity in two French nuclear plants.

After deciding not to participate in the Penly project, in February 2010 GdF Suez sought approval to build an 1100 MWe Areva-MHI Atmea1 reactor at Tricastin or Marcoule in the Rhone valley to operate from about 2020. This sparked union opposition due to the private ownership. It would have been a reference plant for the Areva-Mitsubishi design, providing a base for export sales.

In February 2022 Macron announced plans to build six new reactors by 2050 as part of the country's 'National low carbon strategy' plan, with the first reactor planned to be operational by 2035 (now 2038). A study will also be launched into the possibility of eight additional reactors, supported by the deployment of SMRs. The estimated construction cost for these six reactors was initially put at €50-60 billion, but revised by EDF to €72.8 billion in December 2025. In November 2022 the French Council of Ministers approved a draft legislation intended to streamline the administrative and bureaucratic processes needed to build the new nuclear reactors near, or within, existing sites.

EDF proposes building three pairs of EPR2 reactors, in order, at Penly, Gravelines and Bugey. The Nuclear Policy Council in July 2025 selected Bugey over Tricastin for the third pair, as seismic studies at Tricastin were incomplete; Tricastin remains under study for future reactors. Groundworks began at Penly in 2024, with first concrete expected in 2027.

In December 2025 the Lyon Administrative Court in France annulled amendments to regional and local planning documents that would have enabled construction of two EPR2 units at EDF's Bugey nuclear power plant. The court concluded the environmental impacts of the proposed expansion had not been sufficiently assessed and cited deficiencies in the environmental evaluation process. The court's decision annuls the relevant planning approvals in their entirety effectively suspending the project unless new approvals are secured.

In 2026 France intended to support the construction of the six proposed new reactors at Penly, Gravelines and Bugey with a subsidized loan at a preferential rate covering 60% of the €72.8 billion estimated construction costs, a contract for difference running for 40 years, and some protection from external risks. The European Commission announced it would assess whether this would be in line with state aid rules.

Power reactors under construction and proposed

	Type	MWe gross	Construction start
Penly	EPR2	2x1650	2027?
Gravelines	EPR2	2x1650	-
Bugey	EPR2	2x1650	-

Further nuclear power development

In January 2006 the President announced that the Atomic Energy Commission (CEA)* was to embark upon designing a prototype Generation IV reactor with the aim of achieving operation in 2020, bringing forward the timeline for this by some five years. France has been pursuing three Generation IV technologies: gas-cooled fast reactor, sodium-cooled fast reactor, and very high temperature reactor (gas-cooled). While Areva had been working on the last two types, the main interest in the very high temperature reactors has been in the USA, as well as South Africa and China. CEA interest in the fast reactors is on the basis that they will produce less waste and will better exploit uranium resources, including the 220,000 tonnes of depleted uranium and some reprocessed uranium stockpiled in France.

* Now the Commission of Atomic and Alternative Energy

In December 2006 the government's Atomic Energy Committee decided to proceed with a Generation IV sodium-cooled fast reactor prototype. However in 2019 the CEA announced that it no longer planned to build it. The prototype was to intended to improve the competitiveness and safety of this reactor type, and to demonstrate advanced recycling modes to minimise the ultimate high-level and long-lived waste to be disposed of.

Small modular reactors

TechnicAtome with Naval Group and CEA in France have developed the NP-300 PWR design from submarine power plants and aimed it at export markets for power, heat and desalination. It is a PWR with passive safety systems and could be built for applications of 100 to 300 MWe or more with up to 500,000 m³/day desalination. As of mid-2018, a 570 MWt/170 MWe version was proposed as an SMR to be in a metallic compact containment submerged in water, each module in a separate pool. In September 2019 twin 170 MWe units were proposed to comprise a 340 MWe power plant.

In November 2021 it was announced that first concrete for a demonstration unit was expected in 2030.

The project has evolved into the Nuward SMR, with a revised design of about 400 MWe. In December 2025 a joint early review of the Nuward design by six European nuclear regulators completed its second phase.

Areva and EdF

Areva* was created in 2001 by merging Framatome (later named Areva NP), the nuclear business of Siemens, Cogema (later named Areva NC), and Technicatome (later named Areva TA – the propulsion and research reactor unit). Areva was the only company with a presence in every part of the nuclear fuel cycle. In 2007 it bought the Canadian mining company Uramin for \$2.5 billion, and in 2011 wrote off this investment after concluding that its uranium deposits were of negligible value. Areva's fortunes declined from 2011, with reactor projects in Finland (Olkiluoto 3) and France (Flamanville 3) contributing.

* The name has geographical allusions; it is not an acronym.

In February 2011 the Nuclear Policy Council (*Conseil Politique Nucleaire*, CPN) addressed the rivalry between Areva (almost 90% government-owned) and Electricité de France (EdF, 85% government-owned). This was presumed to have been a factor in losing an important Middle Eastern nuclear power plant contract 14 months earlier. Areva was the world's largest nuclear company and EdF is the largest nuclear electric utility.

The Nuclear Policy Council directed Areva and EdF to put in place a technical and commercial agreement by mid-year for a strategic partnership to improve the design for the European Pressurized Reactor (EPR) and to work together more closely on several fronts domestically. This agreement was signed in July 2011, covering the optimization of Areva's 1650 MWe EPR design that EdF is building at Flamanville 3, improving maintenance and operation of EdF's reactor fleet, and nuclear fuel cycle developments, including new fuels and final disposal of radioactive waste. EdF appeared to have the leading role in this, and particularly in export efforts. The CPN told Areva to spin off its uranium mining into a subsidiary company "as a preliminary step to study strategic and financial scenarios to ensure its development."

In March 2015, Areva announced a two-part strategy to refocus on its core business of nuclear power and return to competitiveness, aiming to make savings of about €1 billion over the next few years after a record loss in 2014 of €4.83 billion. Areva had five operational business units: reactors and services, with engineering and projects (Areva NP); mining; front end; back end; and renewable energies.

Areva SA was 86.52% owned by the government through three entities including the CEA (54.37%), Banque publique d'investissement (3.32%), and Agence des participations de l'État (28.83%). Government-owned EDF held 2.24%, and Kuwait Investment Authority 4.82% (bought for €600 million in December 2010). The balance was held by public investors and employees. (Siemens had a 34% interest in Areva NP until March 2011). In February 2017 Areva SA shareholders approved a reserved capital increase of €2 billion from the French state in order to fund the completion of Olkiluoto 3. This advance was converted into capital in July 2017, so that the state owned about 92% of Areva SA. Also, completion of the sale to EDF would depend on a "favourable outcome" from tests on the Flamanville reactor pressure vessel metallurgy. The ASN provisionally cleared this in June 2017. In July 2017 Areva said that the French state held 67.05% directly and 25.17% through the CEA, total 92.22%. In light of Areva SA's loss of control of NewCo (now Orano – see below) and the sale of New NP (now Framatome – see below), in mid-July 2017 the French state filed a takeover bid at €4.50 each for the shares not held directly or indirectly. This included the Kuwait equity, sold for €83 million.

The financial losses, including €2 billion in 2015, reinforced moves for EDF to take over Areva. Areva said: "Half of this loss of €2 billion

is due to additional provisions for Olkiluoto 3 and half to provisions for restructuring and impairment related to market conditions."

In October 2022 EDF's board of directors approved an offer by the French state to renationalise the company by increasing its shareholding in EDF from 84% to 100% in a deal worth almost €10 billion. The following month, the French Financial Market Authority approved the simplified public tender offer of €12.00 per EDF share and €15.52 per OCEANE (existing shares not already held by the French state). However, in January 2023 the French Financial Markets Authority suspended the state's full takeover until the Paris Court of Appeal rules on a lawsuit brought by minority EDF shareholders who claimed that the price offered to EDF shareholders was too low.

In May 2023 the Court dismissed the claim and the Financial Markets Authority reopened the simplified public tender offer for ten days for the outstanding shares.

EDF was delisted from Euronext Paris on 8 June 2023, completing the renationalization.

Framatome (formerly New NP) and Areva NP

In July 2015 EDF agreed to take a stake of between 51% and 75% of the capital in Areva NP. In November 2016 Areva and EDF signed a contract setting the terms for the sale of New NP*, a 100% subsidiary of Areva NP. EDF announced that the transaction was completed on 2 January 2018, with the French utility acquiring a 75.5% stake. On 4 January 2018, Areva announced that New NP had been renamed Framatome, the name of the former French reactor vendor from which Areva was originally created.

* New NP (now Framatome) was formed from the spin-off of Areva's reactor operations.

Other shareholders in Framatome are 19.5% MHI, 5% Assystem, following the signing of equity agreements in July 2017. During negotiations MHI made it clear that it would share Atmea1 technology only with EDF and not with any other partners in New NP.

Framatome holds all existing assets of Areva related to the design and manufacture of nuclear reactors and equipment, fuel design, fabrication and supply, and services to existing reactors.

Framatome excludes all assets, liabilities and staff related to the completion of the Olkiluoto 3 EPR in Finland, which remain with Areva NP, within the scope of Areva SA, along with some Le Creusot Forge potential liabilities (especially those related to Flamanville 3). Framatome will no longer be tied to a particular reactor design, and long-term operation of all types of reactor will be a major service objective. Annual revenue is about €3 billion.

The six Framatome business units are: large projects; installed base services; fuel design; component manufacturing; I&C (consolidated); and engineering design.

In March 2017 EDF announced a €4 billion share issue, of which the French state would take up €3 billion, reducing its stake in EDF from 85.8% to 83%. The offering of new shares to existing shareholders was oversubscribed by 22%, EDF said. The rights issue was mostly to finance its developments to 2020.

In the course of the sale of New NP to EDF, it was intended that 15% of New NP's capital would be transferred from Areva SA to the new fuel cycle company – Areva NewCo (now Orano) – so that it might maintain a “strategic share” in the reactor business. However, EDF in July 2017 acknowledged discussions between it and Areva “on the conditions for the implementation of the European Commission decision requiring Areva to fully exit New NP at the latest by the end of [the] Areva restructuring plan, planned in 2019.”

In May 2017 EDF announced the creation of the Edvance engineering joint venture with Areva NP, to design and build nuclear islands and control systems for new reactors globally. EDF holds 80%, Areva NP (now Framatome) 20%. This arises from the July 2015 agreement to establish a dedicated company for the design, project management and marketing of new reactors (then called NICE). The aim of this company is to improve the preparation and management of projects, as well as the export offering of the French nuclear industry. EDF said the new company would form part of an “integrated generator/supplier model, which has been tried and tested in several countries.”

Orano (formerly New Areva Holding Co.)

As well as plans to sell most of Areva NP to EDF, in June 2016 Areva announced corporate restructuring through the creation of a new company (“NewCo”) focused on the nuclear fuel cycle apart

from fuel fabrication. The entity, originally named New Areva Holding Co., but renamed Orano on 23 January 2018, was initially a wholly-owned subsidiary of Areva SA set up in July 2016. It combines the Areva Mines, Areva NC, Areva Projects and Areva Business Support entities and their respective subsidiaries.

In November 2016 the Areva shareholders approved the transfer of fuel cycle operations and part of Areva SA's debt to Orano with a €3 billion capital increase. In July 2017, €2.5 billion of the Areva recapitalisation from the government approved by the European Commission (EC) was allocated to Orano. Following the transaction and capital injection – both approved by the EC – the French state would hold, either directly or indirectly, at least two-thirds of the new company's capital, with the remainder held by strategic investors. As noted above, 15% of New NP's capital was intended to be transferred from Areva SA to Orano – so that it could maintain a “strategic share” in the reactor business – though the EC vetoed this and it did not proceed. In January 2017 the EC authorised €1.3 billion rescue aid for Orano from the French state, to be reimbursed by conversion to equity.

In February 2017 Areva announced that Japan Nuclear Fuel Ltd (JNFL) and MHI would each invest €250 million for 5% stakes in Orano. Areva said it has “enjoyed longstanding relations” with both Japanese companies on fuel cycle matters, and that the opportunity remained open for other strategic investors to take equity on the same basis. MHI confirmed that it was also proceeding “toward making a similar minority stake investment in Areva NP.” Both Rosatom and CNNC had earlier expressed interest in Orano.

In February 2018, it was announced that both JNFL and MHI had finalised their investments, giving both organisations a 5% stake in Orano. This announcement marked the end of the restructuring process. The rest of Orano's capital is held by the French state – 45.8% directly, 4.2% through state nuclear agency CEA, and 40% through Areva SA.

The strategic plan of the new organisation was centred on three objectives: to generate more than 30% of its revenue in Asia by 2020; to generate positive net cash flow in 2018; and to ensure more than half of its staff are in service activities in 2020. Orano planned to invest €1.8 billion in modernizing its plants up to 2025.

Balance of Areva

Areva TA, the propulsion and research reactor unit, as well as the renewables businesses, was retained by Areva SA. In March 2017 its 83.5% stake in Areva TA was sold to a consortium comprising the French government's Agency of State Holdings, the CEA, and French naval defence group DCNS. EDF maintains a 9% stake in Areva TA.

Load-following with PWR nuclear plants

Normally base-load generating plants, with high capital cost and low operating cost, are run continuously, since this is the most economic mode. But also it is technically the simplest way, since nuclear and coal-fired plants cannot readily alter power output, compared with gas or hydro plants. The high reliance on nuclear power in France thus poses some technical challenges, since the reactors collectively need to be used in load-following mode. (Since electricity cannot be stored, generation output must be exactly equal to consumption at all times. Any change in demand or generation of electricity at a given point on the transmission network has an instant impact on the entire system). In France, because electricity is cheap relative to other sources (based on imported fossil fuel), electric heating is widespread and a 1°C temperature change in winter means that demand on the grid changes by about 2400 MWe, making it the most temperature-sensitive demand in Europe, adding to the normal challenge of satisfying the balance between supply and demand.

RTE, a subsidiary of EdF, is responsible for operating, maintaining and developing the French electricity transmission network. France has the biggest grid network in Europe, made up of some 100,000 km of high and extra high voltage lines, and 44 cross-border lines, including a DC link to UK. Electricity is transmitted regionally at 400 and 225 kilovolts. Frequency and voltage are controlled from the national control centre, but dispatching of capacity is done regionally. Due to its central geographical position, RTE is a crucial entity in the European electricity market and a critical operator in maintaining its reliability.

All France's nuclear capacity is from PWR units. There are two ways of varying the power output from a PWR: control rods, and boron addition to the primary cooling water. Using normal control rods to reduce power means that there is a portion of the core where neutrons are being absorbed rather than creating fission, and if this is maintained it creates an imbalance in the fuel, with the

lower part of the fuel assemblies being more reactive than the upper parts. Adding boron to the water diminishes the reactivity uniformly, but to reverse the effect the water has to be treated to remove the boron, which is slow and costly, and it creates a radioactive waste.

So to minimize these impacts since the 1980s EdF has used in each PWR reactor some less absorptive 'grey' control rods which weigh less from a neutronic point of view than ordinary control rods and they allow sustained variation in power output. This means that RTE can depend on flexible load following from the nuclear fleet to contribute to regulation in these three respects:

Primary power regulation for system stability (when frequency varies, power must be automatically adjusted by the turbine).

Secondary power regulation related to trading contracts.

Adjusting power in response to demand (decrease from 100% during the day, down to 50% or less during the night, and respond to changes in renewable inputs to the grid, etc.).

PWR plants are very flexible at the beginning of their cycle, with fresh fuel and high reserve reactivity. An EdF reactor can reduce its power from 100% to 30% in 30 minutes. But when the fuel cycle is around 65% through these reactors are less flexible, and they take a rapidly diminishing part in the third, load-following, aspect above. When they are 90% through the fuel cycle, they only take part in frequency regulation, and essentially no power variation is allowed (unless necessary for safety). So at the very end of the cycle, they are run at steady power output and do not regulate or load-follow until the next refueling outage. RTE has continuous oversight of all French plants and determines which plants adjust output in relation to the three considerations above, and by how much.

RTE's real-time picture of the whole French system operating in response to load and against predicted demand shows the total of all inputs. This includes the hydro contribution at peak times, but it is apparent that in a coordinated system the nuclear fleet is capable of a degree of load following, even though the capability of individual units to follow load may be limited.

Plants being built today, e.g. according to European Utility Requirements for LWR Nuclear Power Plants (EUR), have load-following capacity fully built in.

Fuel cycle – front end

France uses some 9700 tonnes of uranium oxide concentrate (8200 tonnes of uranium) per year for its electricity generation. Much of this has come from Orano in Canada and Niger, though Orano lost operational control of its Somair mine in Niger in late 2024 following the country's political changes.

Beyond this, it is self-sufficient and has conversion, enrichment, uranium fuel fabrication and MOX fuel fabrication plants operational (together with reprocessing and a waste management programme). Most fuel cycle activities are carried out by Orano.

Conversion

For over 60 years, Orano has conducted conversion via a two-stage process at its Comurhex plants at the Malvési and Tricastin sites in France. Uranium concentrates (U_3O_8) are converted to uranium tetrafluoride (UF_4) at Malvési. The UF_4 is then transported to Tricastin for final conversion to uranium hexafluoride (UF_6).

In May 2007 Areva NC announced plans for a new conversion project – Comurhex II – expanding and modernizing the facilities at Malvési and Pierrelatte near Tricastin to strengthen its global position in the front end of the fuel cycle. The €610 million project will increase capacity to 15,000 tU/yr, with scope (but no plans) for increase to 21,000 tU/yr. At Malvési near Moussan uranium oxide concentrate is converted to UF_4 powder, and this is sent on to Pierrelatte to produce UF_6 . About 40% of production is on toll basis or exported.

In January 2009 EdF awarded a long-term conversion contract to Areva. On 31 December 2017 Areva announced that it had shut down Comurhex I, with production from the unit to be replaced by Comurhex II. At the start of 2018 Comurhex I had an inventory of three years' worth of sales, from which customers would be supplied between the closure of Comurhex I and the opening of

Comurhex II. The plant was inaugurated in September 2018 and entered commercial operation in December 2018 with a capacity of 7500 tonnes per year. The plant is expected to achieve its full nominal annual production capacity of 15,000 tonnes per year by 2023. Orano stated in September 2018 that EDF is committed to buy about one-third of the total output, with the balance mainly sold under long-term contracts to about 70 utilities in the USA, China, South Korea and several European countries.

Comurhex also converts reprocessed uranium.

Orano has undertaken deconversion of enrichment tails at Pierrelatte since the 1980s. Its 20,000 t/yr W2 plant, the world's largest, produces aqueous HF which is recycled, and the depleted uranium is stored long-term as chemically stable U_3O_8 .

Enrichment

For 33 years this was at Eurodif's 1978 Georges Besse I plant at Tricastin nearby Pierrelatte, with 10.8 million SWU capacity (enough to supply some 81,000 MWe of generating capacity – about one-third more than France's total). Eurodif was by far the largest single electricity consumer in France, using 15 TWh/yr for much of its life. It ran at about half capacity (using about 800 MWe) until mid-2012 and then closed down, as replacement capacity at Georges Besse II reached 1.5 million SWU/yr. The plant delivered more than 200 million SWU, or 35,000 t of enriched product in 33 years.

In 2003 Areva agreed to buy a 50% stake in Urenco's Enrichment Technology Company (ETC), which comprises all its centrifuge R&D, design and manufacturing activities. The deal enabled Areva to use Urenco/ETC technology to replace its inefficient Eurodif gas diffusion enrichment plant at Tricastin. The final agreement after approval by the four governments involved was signed in mid-2006.

The new Georges Besse II enrichment plant at Tricastin was officially opened in December 2010 and commenced commercial operation in April 2011. The €3 billion two-unit plant, which reached full annual capacity of 7.5 million SWU in 2016 (with potential for increase to 11 million SWU), was built and is operated by Orano subsidiary Societe d'Enrichissement du Tricastin (SET). The south plant started construction in 2007, commenced operation in 2011, and reached full capacity of 4.3 million SWU/yr in 2015. Construction of the north plant began in 2009 with first

production in March 2013, and was fully operational at the end of 2016 with 3.2 million SWU/yr capacity. Areva claimed a 60% cost saving in its construction compared with the south plant, due to experience gained and not changing design.

Minority stakes in SET are being offered to customers, and Suez took up 5% in 2008. In March 2009 two Japanese companies, Kansai and Sojitz Corp, jointly took up 2.5%, in June 2009 Korea Hydro & Nuclear Power took a further 2.5%, and in November 2010 Kyushu Electric Power and Tohoku Electric Power each took 1%. The 4.5% Japanese holdings are grouped as Japan France Enrichment Investing Co. (JFEI). EdF as principal customer opted for a long-term contract instead, and in February 2009 it signed a €5 billion long-term enrichment contract with Areva. It runs over 17 years to 2025, corresponding with the amortisation of the new plant. Korea Hydro and Nuclear Power (KHNP) in mid 2007 signed a long-term enrichment supply contract of over €1 billion – described at that time as Areva's largest enrichment contract outside France.

Enrichment will be up to 6% U-235, and reprocessed uranium will only be handled in the second, north unit. A 30% capacity expansion to about 10 million SWU/yr is under way, with civil works more than 70% complete as of December 2025 and initial operations planned for 2028.

The SET plant freed up some 3000 MWe of Tricastin nuclear power plant's capacity for the French grid – over 20 billion kWh/yr (@ 4 c/kWh this is €800 million/yr). The new enrichment plant investment was equivalent to buying new power capacity @ €1000/kW. The GB II plant requires only about 75 MWe (80 kWh/SWU, compared with about 2600 kWh/SWU for GB I).

About 7300 tonnes of **depleted uranium** (DU) tails is produced annually, most of which is stored for future use in Generation IV fast reactors. Only 100-150 tonnes per year is used in MOX. By 2040 this resource is expected to total some 450,000 tonnes of DU.

Enrichment of depleted uranium tails was undertaken in Russia, at Novouralsk and Zelenogorsk. Some 33,000 tonnes of French DU from Areva and EdF was sent to Russia in 128 shipments over 2006-09, and about 3090 t of enriched 'natural' uranium (about 0.7% U-235) was returned. The contracts for this work ended in 2010, and the last shipment was in July 2010 with the returned material shipped by year end. Shipments began again in 2021.

Tails from re-enrichment remain in Russia as the property of the enrichers.

Fuel fabrication is at several Framatome plants in France and Belgium. Significant upgrading of these plants forms part of Framatome's strategy for strengthening its front end facilities. MOX fuel fabrication and use of reprocessed uranium is described below.

A joint venture between Lightbridge and Framatome* was officially launched in January 2018 and assigned the name Enfission. The JV was to develop, fabricate and commercialize fuel assemblies based on metallic fuel technology. However, Lightbridge and Framatome agreed to dissolve the venture in March 2021.

* Areva New NP was renamed Framatome in January 2018 (see [above](#)).

Fuel cycle – back end

France chose the closed fuel cycle at the very beginning of its nuclear programme, involving reprocessing used fuel so as to recover uranium and plutonium for re-use and to reduce the volume of high-level waste for disposal. Recycling allows 30% more energy to be extracted from the original uranium and leads to a great reduction in the amount of waste to be disposed of. Overall the closed fuel cycle cost is assessed as comparable with that for direct disposal of used fuel, and preserves a resource which may become more valuable in the future. Back end services are carried out by Orano. Used fuel storage in pools at reactor sites is relatively brief.

Used fuel from the French reactors and from other countries is sent to Orano's La Hague plant in Normandy for reprocessing. This has the capacity to reprocess up to 1700 tonnes per year of used fuel in the UP2 and UP3 facilities, and had reprocessed 34,000 tonnes as of January 2022. The treatment extracts 99.9% of the plutonium and uranium for recycling, leaving 3% of the used fuel material as high-level waste which is vitrified and stored there for later disposal. Typical input today is 3.7% enriched used fuel from PWR and BWR reactors with burn-up to 45 GWd/t, after cooling for four years.

EdF produces about 1110 tonnes of spent fuel each year. Reprocessing is undertaken a few years after discharge, following some cooling. When reprocessed, the about 1100 tonnes of spent fuel produces 11 tonnes of plutonium and 1045 tonnes of

reprocessed uranium converted into stable oxide for storage. The plutonium is immediately shipped to the 195 t/yr Melox plant near Marcoule for prompt fabrication into about 120 tonnes of mixed-oxide (MOX) fuel, which is used in 24 of EdF's 900 MWe reactors. EDF has made provision to store reprocessed uranium for up to 250 years as a strategic reserve.

At the end of 2019, there were 58 tonnes of civilian plutonium in storage in France, 26% of which was foreign owned.

Used MOX fuel and used RepU fuel is stored pending reprocessing and use of the plutonium in Generation IV fast reactors. These discharges amount to about 130 tonnes per year. Used MOX fuel is not reprocessed at present.

EdF's **recycled uranium (RepU)** is converted in Comurhex plants at Pierrelatte, either to U_3O_8 for interim storage, or to UF_6 for re-enrichment in centrifuge facilities at Seversk in Russia*. About 500 tU of French RepU as UF_6 was sent to JSC Siberian Chemical Combine at Seversk for re-enrichment. The enriched RepU UF_6 from Seversk was then turned into UO_2 fuel in Areva's FBFC Romans plant (capacity 150 t/yr). EdF used it in the Cruas 900 MWe power reactors from the mid-1980s to 2014. The main RepU inventory – 24,000 tonnes at four sites at the end of 2010 but only 16,900 tonnes at the end of 2012 – constitutes a strategic resource, and EdF intends to increase its utilization significantly. The enrichment tails remain at Seversk, as the property of the enricher. After an 11-year hiatus, Orano sent just over 1000 tonnes RepU to Seversk in 2021 to be used in nuclear fuel fabrication in Russia.

* RepU conversion and enrichment requires dedicated facilities due to its specific isotopic composition (presence of even isotopes – notably U-232 and U-236 – the former gives rise to gamma radiation, the latter means higher enrichment is required). It is the reason why the cost of these operations may be higher than for natural uranium. However, taking into account the credit from recycled materials (natural uranium savings), commercial grade RepU fuel is competitive and its cost is more predictable than that of fresh uranium fuel, due to uncertainty about future uranium concentrate prices.

In May 2018 Framatome signed a contract to design, fabricate and supply fuel assemblies using enriched reprocessed uranium to EDF between 2023 and 2032. These will be produced at Framatome's Romans-sur-Isère plant.

Orano has the capacity to produce and market 150 t/year of MOX fuel at its Melox plant for French and foreign customers (though it is licensed for 195 t/yr). In Europe 35 reactors have been loaded with MOX fuel. Contracts for MOX fuel supply were signed in 2006 with Japanese utilities, and a total of six shipments have been made, the most recent in 2017. All these fuel cycle facilities comprise a significant export industry and have been France's major export to Japan.

In addition to LWR fuel, about 5000 tonnes of gas-cooled reactor natural uranium fuel was earlier reprocessed at La Hague, and over 18,000 tonnes was reprocessed at the UP1 plant for such fuel at Marcoule, which closed in 1997.

At the end of 2008 Areva and EdF announced a renewed agreement to reprocess and recycle EdF's used fuel to 2040, thereby securing the future of both La Hague and Melox plants. The 2008 agreement supported Areva's aim to have La Hague operating at 1500 tHM/yr by 2015, instead of two-thirds of that in 2008. Capacity as of 2021 was 1700 tHM/yr.

Under the 2006 Planning Act, each nuclear operator (EDF, Framatome, CEA) manages its waste management and decommissioning fund, which stays inside the company.

Reprocessing developments

France's back-end strategy and industrial developments are to evolve progressively in line with future needs and technological developments. The existing plants at La Hague (commissioned around 1990) have been designed to operate for at least 40 years, so with operational and technical improvements taking place on a continuous basis they are expected to continue operating well beyond 2040, supported by pool storage densification adding 3600 tonnes of capacity and the 'Aval du Futur' programme which plans a successor reprocessing plant (UP4) for operation between 2045 and 2050. In line with progress towards Generation IV plants (reactors and advanced treatment facilities), three main R&D areas include:

The COEX process based on co-extraction and co-precipitation of uranium and plutonium together as well as a pure uranium stream (eliminating any separation of plutonium on its own). This is designed for Generation III recycling plants.

Selective separation of long-lived radionuclides (with a focus on Am and Cm separation) from short-lived fission products based on the optimization of DIAMEX-SANEX processes for their recycling in Generation IV fast neutron reactors with uranium as blanket fuel. This option can also be implemented with a combination of COEX and DIAMEX-SANEX processes.

Group extraction of actinides (GANEX process) as a long term R&D goal for a homogeneous recycling of actinides (i.e. U-Pu plus minor actinides together) in Generation IV fast neutron reactors as driver fuel.

All three processes are to be assessed as they develop, and one or more will be selected for industrial-scale development with the construction of pilot plants. In the longer term the goal is to have integral recycling of uranium, plutonium and minor actinides. In practical terms, a technology – GANEX or similar – will need to be validated for industrial deployment of Gen IV fast reactors about 2040, at which stage the present La Hague plant will be due for replacement.

See also [R&D section](#) below.

Radioactive waste management

Waste disposal is being pursued under France's 1991 Waste Management Act (updated 2006) which established the Agence Nationale pour la gestion des Déchets Radioactifs – [ANDRA](#) – as the National Radioactive Waste Management Agency.

The 2006 revision of the Waste Management Act extended the mandate of the *Commission Nationale d'Evaluation* – [CNE](#) – the National Scientific Assessment Committee, to all wastes. Its role was assessing R&D in three areas concerned with high-level and intermediate-level wastes: deep-geologic disposal, separation and transmutation, and interim storage of nuclear wastes, and this was extended to nuclear materials and all types of waste when CNE2 succeeded the initial CNE in 2006. In April 2007 the government appointed 12 new members to the CNE2 to report on progress in France's waste management R&D across EdF, CEA, ANDRA and the National Centre for Scientific Research. It reports annually.

The 2006 Act was largely in line with recommendations to government from the CNE following 15 years of research. Their

report identified the clay formation at Bure as the best site, but was sceptical of partitioning and transmutation for high-level wastes, and said that used MOX fuel should be stored indefinitely as a plutonium resource for future fast neutron reactors, rather than being recycled now or treated as waste. In a 2010 report CNE2 said that transmutation of minor actinides in fast reactors would add about 10% to power cost, and transmutation of all actinides in an accelerator-driven system (ADS) would add about 20%. Wastes from transmutation reactors will be in interim storage for at least 70 years. In its 2012 report CNE2 noted the great value of plutonium in fast reactors and their role in transmuting long-lived actinides, hence “an experimental reactor and its associated cycle – fuel fabrication and reprocessing – are indispensable” to test “the industrial and economic viability” of that concept while maintaining France’s leadership in civil nuclear energy. In particular the Astrid project would allow “preservation of a range of energy options and ensure France’s energy independence for several centuries.” France’s Astrid project was cancelled in August 2019 (see [R&D section](#) below).

ANDRA sets the direction of research – mainly undertaken at the Meuse/Haute Marne underground rock laboratory in Bure, eastern France, situated in clays. Another laboratory is researching granites. Research is also being undertaken on partitioning and transmutation, and long-term surface storage of wastes following conditioning. Wastes are to be retrievable from the repository. ANDRA publishes a waste inventory every two years and reports to government so that parliament can decide on waste policy.

After strong support in the National Assembly and Senate, the Nuclear Materials and Waste Management Programme Act was passed in June 2006 to apply for 15 years. This formally declared deep geological disposal as the reference solution for high-level and long-lived radioactive waste, and set 2015 as the target date for licensing a repository and 2025 for opening it. It also affirmed the principle of reprocessing used fuel and using recycled plutonium and uranium “in order to reduce the quantity and toxicity” of final waste, and called for construction of a prototype fourth-generation reactor by 2020 to test transmutation of long-lived actinides.

Funds for waste management and decommissioning remain segregated but with the producers, rather than in an external fund. In 2016 the energy minister set the reference cost of the repository project at €25 billion (in 2011 euros), the figure to be updated as construction proceeds. This is slightly higher than EdF

and Areva had provided for in their accounts, so will require adjustments of €800 million and €250 million respectively. ANDRA was expected to lodge a construction licence application in 2017, start construction in 2020, and commence the pilot phase of disposal in 2025, but these dates slipped. Andra submitted its construction licence application in January 2023. In December 2025 the Nuclear Safety and Radiation Protection Authority (*Autorité de sûreté nucléaire et de radioprotection*, ASN) issued a favourable opinion on the safety case. A public inquiry is expected in the second half of 2026, with the target for first receipt of waste packages in 2050. More than half the total cost is expected to be construction, and one-quarter for operation over 100 years.

ANDRA is designing its Bure repository – the Industrial Centre for Geological Disposal (*Centre Industriel de Stockage Géologique*, CIGEO) – to operate at up to 90°C, which it expects to be reached about 20 years after emplacement. In October 2012 CNE2 endorsed the plans for the CIGEO 500-metre deep repository at Bure. Public consultation over May to December 2013 showed that the public was “not opposed in principle” to the project, but wanted a pilot phase demonstration and provision for reversibility. Andra submitted a 'safety options dossier' in April 2016, which was approved by ASN in January 2018. The construction phase is expected to take approximately eight years. It will be designed to take 10,000 cubic metres of HLW, mostly vitrified (from reprocessing 45,000 t used fuel), and 73,000 m³ of long-lived ILW, of which 15,000 m³ is metallic parts from spent fuel.* Vitrified waste canisters will be inserted into long horizontal boreholes 70 cm in diameter lined with steel tubes. In 2015 an amendment to the 2006 Act clarified that for the CIGEO project HLW being 'recoverable' referred to short-term practicality, while 'reversible' meant guaranteeing long-term policy flexibility.

* Initially the CIGEO concept included direct disposal of some categories of used fuel, but the cost implication was considerable due to increased footprint and safeguards management, and the idea was abandoned. Only standard universal canisters will be used, and all fuel will be recycled.

ANDRA's 2023 inventory shows that as of the end of 2021 France then had a total of 1.76 million cubic metres of radioactive waste of which 61% was from power generation, 27% from research, 9% from military, 3% from industry and <1% from medical applications. ANDRA noted that 90% of this volume has a storage and disposal

route through its existing facilities, and once the CIGEO deep geologic repository is in operation it will cater for the balance.

LLW & ILW

ANDRA has the Centre de l'Aube disposal facility for low-level (LLW) and short-lived intermediate-level waste (ILW) near Soulaines in the Aube district, with a capacity of one million cubic metres, over one-third of this so far filled. It opened in 1992 and benefitted from the experience gained at Centre de la Manche. It is operated by an Orano subsidiary. ANDRA also has the Morvilliers facility (CIRES) nearby licensed to hold 650,000 cubic metres of very low-level waste (VLLW), mostly from plant dismantling, in the Aube district around Troyes east of Paris. ANDRA's Centre de la Manche (CSM) facility next to La Hague received 527,000 m³ of low- and short-lived intermediate-level waste from 1969 to 1994, and is now capped with a multi-layer grassed cover.

In June 2008, ANDRA officially invited 3115 communities with favorable geology to consider hosting a facility for the disposal of long-lived LLW (FA-VL, containing radionuclides with half lives of over 30 years). This is 70,000 m³ (18,000 tonnes) of graphite from early gas-cooled reactors and 47,000 m³ of radium-bearing materials from manufacture of catalytic converters and electronic components, as well as wastes from mineral and metal processing that cannot be placed in Andra's low-level waste disposal center in Soulaines. In response, 40 communities put themselves forward for consideration. Preliminary studies completed late in 2008 by ANDRA revealed that two – Auxon and Pars-lès-Chavanges in the Aube district – had suitable rock formations and environments for the disposal of the waste, but after intense lobbying by anti-nuclear groups both withdrew. Investigations are proceeding. A repository is likely to be in clay, about 15 metres below the land surface. Meanwhile ANDRA is building a store for FA-VL waste at its Morvilliers VLLW site.

Financing waste

EdF sets aside 0.14 cents/kWh of nuclear electricity for waste management costs. At the end of 2020, EdF had €24.6 billion provisions in its dedicated back-end fund for France, comprising €11.3 billion for spent fuel management and €13.3 billion for long-term radioactive waste management.

Waste R&D

ANDRA received €75 million in funding under the 'Nuclear of Tomorrow' part of the *Investissements d'Avenir* programme. The funding is being used for various projects:

The CYBER project aims to develop a process for treating rubble using microwave heating to separate components of concrete and find future uses for them. For example, reuse in the nuclear industry in France (e.g. as a material used in disposal facilities), or a recycled use inside or outside the nuclear sector internationally. CYBER aims to avoid the use of mechanical crushing, thereby avoiding the degradation of material characteristics.

ORCADE aims to reduce the volume of VLLW electrical cable waste for disposal by 'stripping' sheathing from cables to recover the metal wire. This may allow it to be reclassified as conventional waste.

The CADET (assisted cavitation for water decontamination) project aims to develop a process for decomposing organic compounds in effluents from nuclear plant decontamination.

Decommissioning

Reactors

Fourteen experimental and power reactors are being decommissioned in France, nine of them first-generation gas-cooled, graphite-moderated types, six being very similar to the UK Magnox type. There are well-developed plans for dismantling these (which have been shut down since 1990 or before) and work is progressing. However, completion awaits the availability of sites for disposing of the intermediate-level wastes and the alpha-contaminated graphite from the early gas-cooled reactors. At least one of these, Marcoule G2, has been fully dismantled.

The other reactors include the 1240 MWe Superphénix (Creys-Malville) fast reactor, the veteran 233 MWe Phénix fast reactor, the 1966 prototype 305 MWe PWR at Chooz, and an experimental 70 MWe GCHWR at Brennilis. A licence was issued for dismantling Brennilis in 2006, and for Chooz A in 2007.

Shutdown Power Reactors in France

Reactor Name	Model	Reactor Type	Reference Unit Power (MWe)	Permanent Shutdown
Chinon A 1	UNGG	GCR	70	1973-04
G 2 (Marcoule)	MAGNOX	GCR	39	1980-02
G 3 (Marcoule)	GCR	GCR	40	1984-06
Chinon A 2	UNGG	GCR	180	1985-06
El 4 (Monts D'Arree)	MONTs-D'ARREE	HWGCR	70	1985-07
St. Laurent A 1	UNGG	GCR	390	1990-04
Chinon A 3	UNGG	GCR	360	1990-06
Chooz A (Ardennes)	CHOOZ-A	PWR	305	1991-10
St. Laurent A 2	UNGG	GCR	465	1992-05
Bugey 1	UNGG	GCR	540	1994-05
Super Phenix	Na-1200	FBR	1200	1998-12
Phenix	PH-250	FBR	130	2010-02
Fessenheim 1	CP0	PWR	880	2020-02
Fessenheim 2	CP0	PWR	880	2020-06

In April 2008 ASN issued a draft policy on decommissioning which proposes that French nuclear installation licensees adopt "immediate dismantling strategies" rather than safe storage followed by much later dismantling. The policy foresees broad public information in connection with the decommissioning process.

In June 2016 EdF told ASN that it was adopting a new strategy for decommissioning the six main GCR reactors at Bugey, Chinon and Saint-Laurent. This will push back the timeline by several decades.

Materials arising from EdF's decommissioning will include: 500 tonnes of long-lived intermediate-level waste, 18,000 tonnes of graphite, 41,000 tonnes of short-lived intermediate-level waste and 105,000 tonnes of very low-level waste.

Fuel cycle

The Eurodif gaseous diffusion enrichment plant at Tricastin closed down in June 2012. A gaseous diffusion rinsing process, designed to recover residual uranium, was completed at the end of 2015. The decommissioning cost is put at €800 million. It is expected to generate 130,000 tonnes of steel and 20,000 tonnes of aluminium that could be recycled, subject to regulatory approval, for use in ANDRA's disposal centres or elsewhere in the industry.

Organization and financing of final decommissioning of the UP1 reprocessing plant at Marcoule was settled in 2004, with the CEA taking it over. The total cost is expected to be some €5.6 billion. The plant was closed in 1997 after 39 years of operation, primarily for military purposes but also taking the spent fuel from EdF's early gas-cooled power reactors. It was operated under a partnership, Codem, with 45% share by each of CEA and EdF and 10% share by Cogema (later Areva NC; now Orano Cycle). EdF and Areva paid CEA €1.5 billion and are clear of further liability.

Financing decommissioning

The total expected cost is periodically re-evaluated, and EDF puts aside an amount related to the total estimated cost, the actualization cost and the expected lifetime of the plants. At the end of 2020 it carried provisions of €20.2 billion for decommissioning and last cores in France, comprising €17.5 billion for decommissioning and €2.7 billion for last cores. It estimates that the total cost (from 2035) will be €75 billion.

In January 2017 a parliamentary committee reported: "The cost of decommissioning is likely to be greater than the provisions," the technical feasibility is "not fully assured" and the dismantling work will take "presumably more time than expected." It questioned the basis of EdF's estimates of €75 billion total cost. EdF responded that it "assumes full responsibility for the technical and financial aspects of dismantling its nuclear plants," and noted that it was currently decommissioning nine reactors, so had a good basis of

experience. It also pointed out that its funds set aside for decommissioning were audited by the Ministry of the Environment, Energy and the Sea the previous month.

Research and development, international

The Atomic Energy Commission (*Commissariat à l'énergie atomique* – CEA) was set up in 1945 and is the public R&D corporation responsible for all aspects of nuclear policy, including R&D. In 2009 it was re-named Commission of Atomic Energy and Alternative Energy (*Commissariat à l'énergie atomique et aux énergies alternatives*, CEA).

The CEA has 4 research reactors of various types and sizes in operation, all started up 1959 to 1980, the largest of these being the 70 MWt Osiris at Saclay, which started up in 1966 for material and fuel testing, and is now being decommissioned. About 34 units dating from 1948 to 1982 are shut down or decommissioning.

In 2004 the US energy secretary signed an agreement with the French Atomic Energy Commission (CEA) to gain access to the Phénix experimental fast neutron reactor for research on nuclear fuels. The US Department of Energy acknowledged that this fast neutron "capability no longer exists in the USA." The US research with Phénix irradiated fuel loaded with various actinides under constant conditions to help identify what kind of fuel might be best for possible future waste transmutation systems.

In mid-2006 the CEA signed a four-year €3.8 billion R&D contract with the government, including development of two types of fast neutron reactors which are essentially Generation IV designs: an improved version of the sodium-cooled type (SFR) which already has 45 reactor-years operational experience in France, and an innovative gas-cooled type. Both would have fuel recycling. The CEA sought support under the EC's European Sustainable Nuclear Industrial Initiative and partnerships with Japan and China to develop SFR which will have great flexibility in breeding ratios. It noted that China and India are aiming for high breeding ratios to produce enough plutonium to crank up a major push into fast reactors.

The National Scientific Evaluation Committee (CNE) in mid-2009 said that the sodium-cooled model, **Astrid** (Advanced Sodium Technological Reactor for Industrial Demonstration), should be a

high priority in R&D on account of its actinide-burning potential. It was initially envisaged as a 600 MWe prototype of a commercial series of 1500 MWe SFR reactors which were planned to be deployed from about 2050. These would consume the plutonium in used MOX fuel and utilise the half million tonnes of depleted uranium (DU) that France will have by 2050. Astrid would have high fuel burn-up, including minor actinides in the fuel elements, and while the MOX fuel would be broadly similar to that in PWRs, it would have 25-35% plutonium and negative void reactivity in the core. It would use an intermediate sodium coolant loop, though whether the tertiary coolant is water/steam or gas was an open question. Over 2014-16 experiments with Brayton cycle gas turbine technology driven by nitrogen were carried out with the CEA. Four independent heat exchanger loops were likely, and it was to be designed to reduce the probability and consequences of severe accidents to an extent that is not now done with FNRs. Astrid is called a 'self-generating' fast reactor rather than a breeder in order to demonstrate low net plutonium production. CEA planned to build it at Marcoule, but the project was cancelled in August 2019. About €735 million had been spent on the project.

Earlier in September 2010 the government had confirmed its support, and €651.6 million funding to 2017, for a 600 MWe Astrid prototype. (A further €350 million was later approved to 2020.) In December 2012 it approved moving to the design phase, with a final decision on construction to be made in 2019. The six-year conceptual design was finished in 2015. The basic design phase ran to 2019, with 14 industrial partners.

The Astrid programme included development of the reactor itself and associated fuel cycle facilities: a dedicated MOX fuel fabrication line (AFC) that was planned to be built about 2017 and a pilot reprocessing plant for used Astrid fuel (ATC) about 2023.

A major tripartite France-US-Japan accord on developing fast reactors was signed in October 2010, and some Astrid safety and performance features were checked by the Idaho National Laboratory in USA. In May 2014 Japan committed to support Astrid development, and in August 2014 JAEA, Mitsubishi Heavy Industries and Mitsubishi FBR Systems concluded an agreement with the CEA and Areva to progress cooperation on Astrid. In 2015 JAEA with MHI-MFBR became the second largest contributor to the program, after Areva NP. In March 2017 the Japanese partnership was strongly reaffirmed after Japan decided to decommission its Monju FNR.

In June 2018 the French government stated that Astrid capacity was to be scaled down from the initially planned 600 MWe to between 100 and 200 MWe to reduce construction costs and also due to development of a commercial fast reactor no longer being a high priority. Following the decision, Toshiba said that the smaller Astrid would be a step back for Japan's fast reactor development process, possibly forcing the country to build its own larger demonstration reactor in Japan rather than rely on Astrid.

CNE is a high-level scientists' panel set up under the 1991 nuclear waste management act and charged with reviewing the research and development programs of the organizations responsible for nuclear energy, research and waste. The CNE expressed a clear preference for the concept of heterogeneous recycling of minor actinides, called CCAM. In that process, minor actinides are separated out from used fuel in an advanced-technology reprocessing plant and then incorporated into blanket assemblies which are placed around the core of a future fast reactor. Such blanket assemblies could contain 20% minor actinides or more, dispersed in a uranium oxide matrix. (In homogeneous recycling, the actinides are incorporated into the actual fuel.)

Allegro was the second line of French-led FNR development – also a Euratom project under the European Sustainable Nuclear Industrial Initiative (ESNII). It is now the demonstration project for the reference gas-cooled fast reactor (GFR), one of the six or seven designs promoted by the Generation IV International Forum. A 50-100 MWt experimental version would initially have a MOX or UO₂ driver core with outlet temperature of 530 °C, then an intermediate core with up to six refractory mixed carbide fuel assemblies, then a final refractory carbide core with 850 °C outlet temperature. Two primary helium circuits connect to secondary circuits with gas or pressurized water. Three decay heat removal loops are integrated in a pressurized guard vessel. The Czech Republic, Hungary, Slovakia and Poland (i.e. the Visegrád countries) made a joint proposal to host the project, with French CEA support, and the V4G4 Centre of Excellence was set up in Slovakia in 2015 (V4G4 = Visegrád 4, Generation IV). The project's preparatory phase is planned to 2026. Further details are in the information page on [Fast Neutron Reactors](#).

In 2015, CEA's nuclear research centres in Saclay and Cadarache became the first to be designated international research hubs under the International Centres based on Research Reactors (ICERR) programme launched by the International Atomic Energy Agency

(IAEA) the previous year. Related to this, the CEA has signed agreements with Jordan, Morocco, Tunisia, Algeria, Slovenia and Indonesia. The ICERR program allows participating research reactors in its framework to coordinate and rationalise their offer of facilities, resources and services to interested IAEA member states.

In March 2007 the CEA started construction of a 100 MWt materials testing reactor at Cadarache to replace Osiris. The **Jules Horowitz reactor (JHR)** with twice the neutron flux of Osiris is the first such unit to be built for several decades, and has been identified by the EU as a key infrastructure facility to support nuclear power development, as well as producing radioisotopes and irradiating silicon for high-performance electronic use. The €500 million cost is being financed by a consortium including CEA (50%), EdF (20%), Framatome (10%) and EU research institutes (20%). Since the anticipated planned high-density U-Mo fuel is not likely to be ready in time, it will start up on uranium silicide fuel enriched to 27%. Civil engineering work for the reactor building was completed in March 2017. Areva (now Framatome) designed and is building it.

Also at Cadarache, Areva TA with DCNS built a test version of its Réacteur d'essais à terre (**RES**), a land-based equivalent of its K15 naval reactor of 150 MWt, running on low-enriched fuel. It reached maximum power for the first time in March 2021. It has also designed the **NP-300** reactor based on these, able to be built in sizes up to about 300 MWe.

In January 2011 DCNS announced the **Flexblue** submerged nuclear power plant concept, developed in collaboration with Areva, EdF and CEA. A 50 to 250 MWe nuclear power system (reactor, steam generators and turbine-generator) would be housed in a submerged 12,000 tonne cylinder about 100 metres long and 12-15 metres diameter, offshore at about 60-100 m depth. DCNS is a state-owned naval defence group formed in 2007 from the merger of DCN shipyard and Thales SA, and makes nuclear submarines and surface ships. It has built 18 nuclear reactors for the French navy and is involved in building the RES test reactor and some components for EPR reactors.

In relation to introduction of Generation IV reactors, the CEA is investigating several fuel cycle strategies:

Optimizing uranium and plutonium recycling from present and EPR reactors, then co-management of U&Pu and possibly Np in

Gen IV fast reactors.

Recycling these with a low proportion of minor actinides (eg 3% MA) in driver fuels of Gen IV fast reactors.

Recycling (in about one third of France's reactors) with up to 30% of minor actinides in MOX blanket assemblies of Gen IV fast reactors.

CEA is part of a project under the Generation IV International Forum investigating the use of actinide-laden fuel assemblies in fast reactors – The Global Actinide Cycle International Demonstration (GACID). See [Generation IV Nuclear Reactors](#) paper.

Nuclear technology exports

The well-established 900 MWe PWR design was sold to several export markets: Iran (2), South Africa (2) and South Korea (2) and China (4). There are two 900 MWe French reactors operating at Koeberg, near Cape Town in South Africa, two at Hanul/Ulchin in South Korea and four at Daya Bay/Ling Ao in China, near Hong Kong. The deal with Iran collapsed politically in 1979 and the engineering components retained in France were used at Gravelines. China's CPR-1000 design is based on the four French M310 units.

Framatome in conjunction with Siemens in Germany then developed the European Pressurised Water Reactor (EPR), based on the French N4 and the German Konvoi types, to meet the European Utility Requirements and also the US EPRI Utility Requirements. This received French design approval in 2004. Framatome has sold eight EPRs – to Finland (1), France (1), China (2) and the UK (4). In 2009 Areva with GDF-Suez and Total lost a bid to build four EPRs near Abu Dhabi in the UAE.

Export sales and prospects for French nuclear power plants

Country	Plant	Type	Initial est. cost	Company	Status, financing
Iran	Darkhovin 1&2	M310	\$2 billion	Framatome	Cancelled in 1979
South Africa	Koeberg 1&2	M310		Framatome	Commissioned 1984-85

Country	Plant	Type	Initial est. cost	Company	Status, financing
South Korea	Hanul/Ulchin 1&2	M310		Framatome	Commercial operation 1988-89
China	Daya Bay	M310		Framatome	Commercial operation 1994
China	Ling Ao	M310		Framatome	Commercial operation 2002
Finland	Olkiluoto 3	EPR		Areva NP	Commercial operation 2023
China	Taishan 1&2	EPR		Areva NP	Commissioned 2018-19
Turkey	Sinop 1-4	Atmea1	\$22 billion	MHI-EDF	Stalled; MHI-EDF consortium withdrew
UK	Hinkley Point C 1&2	EPR	£22.5 billion	Framatome	Under construction
UK	Sizewell C 1&2	EPR		Framatome	Approved; financial close November 2025

After Areva lost its bid to build EPRs in the UAE, the Nuclear Policy Council (CPN) in 2011 called on Areva, EdF, GdF-Suez (now Engie) and "other stakeholders" to strengthen their collaboration on the Atmea1 power reactor. This is a medium-sized (1100 MWe) Generation III design being developed under a 2006 joint venture by Areva NP and Mitsubishi Heavy Industries. The reactor is intended for marketing primarily to countries embarking upon nuclear power programs, although CPN said that construction of an initial Atmea1 in France, as proposed by GdF Suez, would be considered. In addition, the Ministry of Energy would lead a working group to look into the technical, legal and economic aspects of small (100-300 MWe) reactor designs. In May 2013 the Turkish government accepted a proposal from a consortium led by Mitsubishi Heavy Industries (MHI) and Areva, with Itochu, for four Atmea1 reactors at Sinop, at a cost of some \$22 billion, but that project has not yet proceeded.

The **Nuclear Sector Strategy Committee** (CSFN) was set up in February 2011 by the CPN and comprises representatives of 80 companies and industry organizations. It is headed up by EdF. It is an expression of French determination to regain a major role in nuclear exports through "patriotic solidarity". A new trade association, Gifen, was established.

These 2011 policy developments incorporated the role of the **Agence France Nucleaire International (AFNI)**, created in May 2008 under the CEA to provide a vehicle for international assistance. Its purpose is to help to set up structures and systems to enable the establishment of civil nuclear programmes in countries wanting to develop them and to draw on all of the country's expertise in this. It was dissolved in 2019.

Regulation, safety & non-proliferation

The General Directorate for Nuclear Safety and Radiological Protection (DGSNR) was set up in 2002 by merging the Directorate for Nuclear Installation Safety (DSIN) with the Office for Protection against Ionising Radiation (OPRI) to integrate the regulatory functions and to "draft and implement government policy."

In 2006 the new Nuclear Safety Authority (*Autorité de sûreté nucléaire* – [ASN](#)), an independent body with five commissioners – became the regulatory authority responsible for nuclear safety and radiological protection, taking over these functions from the DGSNR, and reporting to the Ministers of Environment, Industry & Health. However, its major licensing decisions still require government approval.

Research was undertaken by the IRSN – the Institute for Radiological Protection & Nuclear Safety – also set up in 2002 from two older bodies. The IRSN was the main technical support body for the ASN.

In February 2023 the Ministry of Energy Transition announced that ASN and IRSN were to be merged which they believe will "strengthen the independence of nuclear safety control, within a single independent safety unit, in order to guarantee a high level of safety requirements."

The merger was enacted into law in May 2024, and the new *Autorité de sûreté nucléaire et de radioprotection* (Nuclear Safety and Radiation Protection Authority, ASNR) became operational on 1 January 2025.

There have been two INES Level 4 accidents at French nuclear plants, both involving the St Laurent A gas-cooled graphite reactors. In October 1969, soon after commissioning, about 50 kg of fuel melted in unit 1, and in March 1980 some annealing occurred in the graphite of unit 2, causing a brief heat excursion. On each

occasion the reactor was repaired, and the two were eventually taken out of service in 1990 and 1992.

The French Nuclear Energy Society (SFEN) is a professional association.

Non-proliferation

France is a party to the Nuclear Non-Proliferation Treaty (NPT) which it ratified in 1992 as a nuclear weapons state. Euratom safeguards apply in France and cover all civil nuclear facilities and materials.

In addition, IAEA applies its safeguards activities in accordance with the trilateral "voluntary offer" agreement between France, Euratom and the IAEA which entered into force in 1981.

France undertook nuclear weapons tests 1960-95 and ceased production of weapons-grade fissile materials in 1996. Since then it has ratified the Comprehensive Test Ban Treaty.

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